

BT 605 Midterm Past Papers

1. Which is not patient right? _____ **COMMON COLD**
2. Genetic diagnosis referred to as _____ DNA testing-diagnosis of genetic diseases-determine a child parentage **ALL**
3. Designer babies are made through _____ **IN VITRO FERTILIZATION**
4. American Biological Safety Association founded in _____ **1984**
5. _____ transplanted the nucleus of a tadpole intestinal cell into an enucleated frog egg. _____ **JOHN GURDON**
6. systemic search for person with specific genotype, individual or group show a risk of disease called _

GENETIC SCREENING

7. who introduced bioethics _____ **POTTER AND CALLAHAN**
8. There are _____ codes of nurses _____ **4 (FOUR)**
9. Nano robots used for treatment of _____ **CANCER/ CANCER PATIENT**
10. ethics is related to _____ **scientific research**
11. biohazard symbol discovered by _____ **CHARLES BALDWIN**
12. Creation of sever sibling enable the identification of tissue match for an older sibling suffer from life threatening disease
13. hESCs are used to _____ **GENERATE ORGANS**
14. Meta ethics is not (observational, theoretical, experimental, none) lec 3 _____ **Experimental**
15. For which disease, personal health info should be disclose to family (cancer, aids, tuberculosis, none) lec25 **Tuberculosis**
16. Religious thinkers believe that genetic screening is a sort of playing with (human, DNA, God, religion) lec 42 **God**
17. Advantage took place in 21st century in all of the following except (technology, education, administration, none) lec 6 **none**
18. Use of viruses in genetic modification as vector increase the risk of _____ **CANCER INDUCTION**
19. The act which is not responding to trafficking _ (use of force, threat, education, abuse of power) _

EDUCATION

20. Creation of sever sibling enable the identification of tissue match for an older sibling suffer from life threatening disease __ **(HFE 1990 ACT)**

VU Medical Zone Effort By Amaan Khan

21. Cybernetics derived from Greek word _____ **GOVERNANCE**
22. According to Islam soul is added to the embryo after _____ **40 DAYS**
23. Neurodegenerative disease occurs at the age of _____ **40 YEARS**
24. Living will includes _____ Treatment Plan (lec24)
25. Which one is the cause of global warming _____ **INCREASE IN CO₂**
26. The idea of contaminant was used to prevent from the spread of communism by _____ **USA COLD WAR**
- POLICY**
27. One of the following is incorrect, which one option is it? (a. physician has responsibilities, b. patients has rights, c. **If patient is mentally retarded than has no right at all,** d. Patient should provide information to make decisions) Lec 19 **ans;C**
28. Which one is used to purify enzymes and sugar solution (moist heat, dry heat, membrane filter, autoclave) Lec 72 **membrane filter**
29. All metals are toxic except **(gold)** lec 12
30. Which one is not an intrinsic value _____ Value of birds/green, places/religious belief _____ **(other than these)**
31. Who introduced bioethics...a.mike and fraud B. Richard and Thomas (c) Thomas and james (d) **Donald land Car**
32. All of these are benefits of using animals in biomedical research but one of these is not one
- a. benefit of animal research is enormous. B. **ANIMAL CANNOT COMPLAIN ABOUT SUFFERING**
- b. Reduced pain experienced by them d. Good experience reduce the number of animals
33. Patients don't have right ...ANS. **to terminate his life**
34. Biosafety depends upon...ANS. **strict guidelines to follow.**
- prevention of large-scale loss of biological integrity, prevention mechanisms, conduction of regular reviews - biosafety in laboratory settings, strict guidelines _____ **All**
35. spiritual treatments like yoga are useful for the patients of... _____ cancer,tb, **psychologically disstress,** lupus erythematosus.
36. Animals are used recreationally except _____ (sports, food, leisure, fashion)_____ **FOOD**
37. which is not code of nurses _____ **NURSES AND FAMILY**

38. All are the sources of increase in co2 level except ____ **USE OF SOLAR PANELS TO TRAP ENERGY**
39. ____are making ethical values since 300 years ago (A scientists -B philosopers -C ethoists -D **all**)
40. Cybernetics is ____ (**A only mechanical** B only biological C only philosphical D **none**) (because its related to all)*
41. Gene patenting is an ____ **INVENTION**
42. which is keep at nearest and earliest place in lab so that every person can see it ____ (**A first aid kit** B biological chemical C poisionious chemical D flammable liquids)
43. Harvesting of human embryo violate ____ **second duties** **respect the value of human life** (lec 51)
44. To be subject of a patent, an object must be a/an ____ **INVENTION NOT DISCOVERY**
45. Which one is not required in personal health information ____ **PERSON HEALTH INFORMATION OF FAMILY**
46. In Xeno-transplantations means we can transplant cells or organs ____ **From specie to another specie**
47. The degradation of following cells releases the allergy causing chemicals such as histamine. T-cells, b-cells, **mast cells**, and monocytes
48. DNA consists of all except ____ **RIBOSE**
49. Designing babies can, the increase of human life span up to ____ **30 YEARS**
50. The scientific research is related to with ____ **ethics**
51. Dolly sheep suffered ____ **RESPIRATORY DISORDER**
52. Which one of the following is applied on the single on a local level ____ **BIOACCUMULATION**
53. Which of the following is a climax forest ____ **TROPICAL RAIN FOREST**
54. Animals that are used in walking and draughting are called.... **beast of burden**
55. The word science came from ____ **LATIN "SCIO"**
56. Nanorobots when injected the treatment of cancer patient **seek only cancerous cells**
57. Cybernetics is a Greek word which means ____ **GOVERNANCE**
58. God had spoken through his ____ commandment. ____ **TEN (10)**
59. ____used in refrigerator as a coolant and destroy ozone layer ____ **AEROSOL SPRAYS**

60. Human body containscells.____ **100 TRILLION**
61. Our DNA is ____ similar to chimpanzees. ____ **98%**
62. HGP stands for ____ **HUMAN GENOME PROJECT**
63. Stem cells research is acceptable due to ____ benefits ____ **THERAPEUTIC**
64. Germ weapons, disease producing infectious agents used against humans, animal or plants
____ **BIOWEAPONS**
65. In 1850s ____ forces developed containment strategy. ____ **Anti-slavery**
66. Removal of microbes including bacterial spores called ____ **STERILIZATION**
67. Meta ethics means____ **THEORETICAL**
68. According to safety protocol taking excess amount of chemical in laboratory can result in the ____ **fumes**
production
69. "Suddenly overwhelming and event occurring "Disaster definition is done by *66*
a)Laymans b)Academics **c)Internationa Red cross community** d)United nations
70. Factors affected development of Bioethics.excepted
a)advances in biomedical science b)environmental ethical concerns
c)ethical issues related to artificial intelligence d)ethical issues related to animals
71. A patient cannot refuse the treatment.if
a)He/she having an infectious disease
b)He/she having an extreme pain due to cancer.i.e
Chemotherapy c)it is small antibiotic with little side effects
d)it is blood transfusion with minimal blood risks
72.prohibited mouth pipetting.
a)UN **b)WHO** c)WWN
73. Harvest of the fetus violates the
a) respect the value of human life b) US Law c)Both duties d) Has rights because it is in a viable form

74. According to lab safety protocols, taking the excess chemical is

- a) loss to the lab b) contamination of lab by fumes c) risk the fire of colleagues d) ill reputation to the scientists

75. special approval is required for the use of

- a) toxins b) Animals c) Infectious agents d) Radioactive material

76. Which is not a physical method to kill the bacteria (sterilized).

- 1) Heat. 2) Radiation. 3) Filtration. 4) Chlorination

77. Cybernetics derived from.....word _____ **GREEKWORD GOVERNANCE**

Advantage took place in 21st century in all of the following except

- a. Technology b. Education. C .adminstration. d. None of above

78. what is the important advise for the reviewers of research grant proposed in the code of ethic.....not

allow use of manuscript in improper way

79. Regulatory system in cybernetics can be _____ **Structures and functions**

80. Unrequired event which occur suddenly is called _____ **DISASTERS**

81. Practice of using bio weapon in war is..... **ancient**

82. person who has genetically carrier

a) have carrier of gene but don't express in next generation.

B) have not carrier of gene but do express in next generation c) have carrier of gene and gain disease but don't express n .generation d) have carrier of gene but don't gain disease, but express in next generation.

83. Plagrism is ?

- A) summarize dishonesty B) dishonesty with coworkers C) honesty with coworkers D) dishonesty with family members

84. Accident report will be written within _____ **FIVE DAYS**

85. A person purchases a glass ware should be noted _____ **BOROSILICATE**

86. Nano robots are used for the disease is _____ **CANCER**

87. embryo has.... **own rights**

88. earth came into being approximately... 3.5 billion years ago
89. approximately animals are used in USA for research_____ 26 million
90. in living will patient talk about _____ treatment plan
91. surgical delivery is recommended in case of . Alzheimer disease.
92. dietary DNA is degraded by_____ enzyme
93. in xenotransplantation graft of... (organ/tissue transplantation, designer babies)
94. while purchasing lab glassware it should be noted the glass is.. boreosilicasat
95. microbes unlikely to cause disease are include .. WHO Group I
1. which one of following is not disaster same as patenting gene (not confirm)
2. viruses is derived from... toxin
3. one the following will not define according to aristotle
- (a)virtues of happiness and sadness (b)ethics and moral thinking with concept of sin
- (c)ethics of particular individuals. (d)feeling humane behavior
- 4.Earth come into beings appear...
- (a)2.5 billion years ago (b)3.5 billion years ago (c)4.5 billion years ago (d)1.5 billion years ago
4. Patience advanced direct is _____ legal document
5. the ideal containmnt was used by.....as miltary stratrgy to contain spread of commus...
- (a)germny (b)japan (c)USSR (d)USA
6. saver fever tomato (1994)is _____ genetically modified .
7. practic of bio weapons is the warfare....
- (a)common (b)ancient (c)modern (d)effective to destroy infrastructure
8. After spillage priority must be given to
- (Clean up spillage,first aid,making report,evaluation of damages)
9. which is infectious agent??
10. hEScs obtained from_____ blastocyst stage
11. which famous philosopher _____ said people have their own rights. peter

12. sometimes the knowledge that one is certain to suffer a serious and distressing condition is a burden too heavy to bear so
- A) genetic counselling is advised.
13. favr savr tomato 1994 is genetically modified tomato and has.....
- A, longer life cycle, no alien gene, improve taste , all
14. in which of the following patient has right to withhold information to his/her spouse given by physician, A, brain tumor, tb, hepatitis, aids.
15. spiritual treatments like yoga are useful for the patients of...
- A, cancer, tb, psychologically distress, lupus erythematosus.
16. animals are used recreationally except
- A, sports, food, leisure, fashion.
17. meta ethics is not
- A, observational, theoretical, experimental, none
18. Earth came into being approximately _____ 3.5 billions years ago
19. In living will patient talk about _____ treatment plan
20. There are aspects of codes for nurses to deal with their professional life except. _____
(nurses and people, nurses and practice, nurses and profession, nurses and co-workers)
21. In xenotransplantation graft of _____ (organ/tissue transplantation, designer babies)
22. While purchasing glassware it should be noted that glass is _____ borosilicate
23. Microbes unlikely to cause diseases include _____ WHO Group I
24. Virus is derived from _____ toxin
25. Which of following is not disaster same as patenting gene _____
26. hESCs derived embryo from _____ blastocyst stage
27. nano robots used for treating _____ cancer patients
28. accident report made within _____ five days
29. 70 years old man eat food _____ (30 goats, 600 hens, 3 cows)
30. Patent of gene must be Not discovery _____ invention
31. Which famous philosopher said people has their own rights _____ peter
32. After spillage priority must be given to _____ first aid to worker injury
33. One of the following not define according to Aristotle _____ ethics and moral thinking
with concept of sin
34. Who introduce the bio ethics _____ potter and callahns

35. Which is fall climax forest _____ **tropical rain forest**
36. There are Codes of nurses _____ **4**
37. Patient advanced directive is _____ **legal document**
38. The ideal contaminant was used to prevent the spread of communism abroad by _____
USA-Cold War policy
39. Favv Savr tomato genetically modified tomato _____ **(1994)**
40. Practice of bio weapons in ware fare is _____ **ancient**
41. De-granulating of release allergy causing chemicals. _____ **mast cells**
42. Health care is not associated/related to _____, technology
research education/administration /communication **,none of these**
43. Gene therapy is not acceptable for _____ **leukemia**
44. All are example of contamination zone except. _____ **option A** (B. sealed rooms, C. sealed containers, D. personal suits)
45. Staphylococcal enterotoxin B is an _____ **infectious agent**
46. hESCs is used to _____ **generate organ**
47. Following is not component of ACT _____ **abuse of power**
48. Who was the philosopher says that every animal has its own right. _____ **peter**
1. DNA consist if all accept ne _____ **Phosphate.**
 2. Surgical delivery use for _____ **Alzheimer disease.**
 3. _____ derived from embryo 5-7 days old before implantation. _____ **hESCs**
 4. Increases human life span up to _____ **30days.**
 5. _____ creation of sever sibling. Enable the identification of tissue match for an older sibling, suffer from life threatening disease. _____ **HFE 1990 Act.**
 6. Required special approval for _____ animal, all _____ **infectious agent.**
 7. Dietary DNA is degraded by _____ **enzyme**
 8. Cybernetics derived from Greek word _____ **governance.**
 9. Playgrism is _____ **without honesty.**
 10. Not a part of ecosystem. ____ (light..air..water..soil)____ **light**
 11. Is not a factor of development _____
 12. Science is _____ observation and experimentation
 13. Threat to science is result are experimentally built in science but _____ socially constructed.
 14. Ethics is related to science _____ philosophy and religion.
 15. Practical meaning related to _____ normative ethics.
 16. Health care is associated/related to _____ technology.
 17. Natural world is related to, except to _____ human
 18. In USA number of animal use in research _____ **26 million**

19. According to Islam soul is added to the embryo_____ 40days
20. Carrier of gene is a person who _____ do not have symptoms but carries disease to their offspring.
21. Gene patenting is related to _____ - gene creation/ created gene.
22. Microbes cause disease _____ WHO risk group.
23. Neurodegenerative incur in age 40 , 50 60
24. Embryo have emotional , psychological , right to treat as human
25. All metal poison except Gold
26. To diagnosis the fetal tissue which are required for diagnosis Allantoic fluid , amniotic , chorion
27. Type of allergy when used the fur of animal..... Hypersensitivity type 1 , 2 , 3 ,4
28. Dolly sheep suffered lack to walk, respiratory disorder etc
29. Reducing the folic acid Anemia , thalassaemia
30. if the condition of the patient is critical how much news give to the patient
31. if a mother refuses the treatment what is the response of the physician
32. Science is derived from a _____ word _____ latin
33. Virtues is given by _____ philosophers _____ greek
34. The environment and living forms are involved in _____ -values
35. Which animal is now produced for fur? Mink
36. Aristotle idea rejected by _____ Thomas brooke
37. Patient has a right to _____ receive information, make decisions, confidentiality, Continuity, availability of health care _____ receive information
38. Informed consent _____ is legal and ethical right of the patient, permission before getting the health care.
39. Human stem cells are obtained from _____ cells. The serious brain fatal disease is caused at the age of _____.
40. Human stem cells are obtained _____ stage
41. Neurodegenerative incur in age 40 , 50 60
42. Bio hazards symbol was discovered by-----

MCQs:

1. DNA consists of all accept _____. Phosphate.
2. Surgical delivery use for _____ cephalus disease/ cephalic disorder. 27
3. _____ derived from embryo 5-7 days old before implantation. _____ hESCs 52
4. Increases human life span up to _____ 30years. 53
5. _____ creation of sever sibling. Enable the identification of tissue match for an older sibling, suffer from life threatening disease. _____ HFE 1990 Act. 56

6. Required special approval for infectious agent. _____ (animal, plant, human, **all**). 69
7. 98% Dietary DNA is degraded by _____. **enzyme** 30
8. Cybernetics derived from Greek word _____ **governance**. 33
9. Playgrism is _____ **without honesty**. (not clear and confirm)
10. Not a part of ecosystem. ____ (light..air..water..soil)_____ **light**. 8
11. Science is related to _____ **observation and experimentation**.
12. Threat to science is result are experimentally built in science but _____ **socially constructed**. 1
13. Ethics is related to science _____ **philosophy and religion**. 3
14. Practical meaning related to _____ **applied ethics**. 3
15. Health care is associated/related to _____ **technology**. 6
16. Environment is a natural world is related to all aspects except to _____ **human**. 8
17. In USA number of animal use in research _____ **26 million**. 15
18. According to Islam soul is added to the embryo _____ **40days**. 51
19. Carrier of gene is a person who _____ **do not have symptoms but carries disease to their offspring**. 60
20. Gene patenting is related to _____ **gene creation/ created gene**. 46
21. Microbes cause disease _____ **WHO risk group 2**. 73
22. Neurodegenerative incurs in age 40. 60
23. Embryo have _____, right to treat as human. _____ emotional , psychological. 51
24. All metal poison except _____ Gold. 12
25. To diagnosis the fetal tissue which are required for diagnosis _____ amniotic cells.
26. Which of allergy occur by the fur of animal _____ type 1 Hypersensitivity. 77
27. Dolly sheep suffered _____ **back to walk**, respiratory disorder. 50
28. Reducing the ____ disease by adding folic acid in wheat. _____ Anemia. 34
29. if the condition of the patient is critical how much news give to the patient. _____ **70-80%**. 22
30. if a mother refuses the treatment what is the response of the physician. _____ **Should not force her**. 27
31. Science is derived from a _____ word _____ **latin** 1
32. Virtues is given by _____ philosophers _____ **greek** 7
33. The environment and living forms are involved in _____ **intrinsic values**. 9
34. Which animal is now produced for fur? **Mink**. 16
35. Aristotle idea rejected by _____ **Thomas**. 4
36. Patient has a right to _____ receive information, make decisions, confidentiality, Continuity. **All**. 19
37. _____ is legal and ethical right of the patient, permission before getting the health care. ____ **Informed consent**. 23

38. Human stem cells are obtained from _____ cells. Embryo. 52
39. The serious brain fatal disease is caused at the age of _____ 40. 60
40. Bio hazards symbol was discovered in 1966 by _____ Charles Baldwin. 70

BT605 BIOSAFETY AND BIOETHICS MIDTERM PAST MCQS AND SUBJECTIVE

MCQs:

1. DNA consists of all except . **Phosphate.**
2. Surgical delivery use for cephalus disease/ cephalic disorder.
3. Derived from embryo 5-7 days old before implantation. **hESCs**
4. Increases human life span up to **30years.**
5. Creation of sever sibling. Enable the identification of tissue match for an older sibling. suffer from life threatening disease. **HFE 1990 Act.**
6. Required special approval for infectious agent. (animal, plant, human, all). 7. Dietary DNA is degraded by . enzyme
8. Cybernetics derived from Greek word **governance.**
9. Playgrism is **without honesty.**
10. Not a part of ecosystem. (light..air..water..soil) **light.**
11. Science is related to **observation and experimentation**
12. Threat to science is result are experimentally built in science but **socially constructed.**
13. Ethics is related to science **philosophy and religion.**
14. Practical meaning related to **applied ethics.**
15. Health care is associated/related to **technology.**
16. Natural world is related to, except **to human.**
17. In USA number of animal use in **research 26 million.**
18. According to Islam soul is added to the embryo **40days.**
19. Carrier of gene is a person who **do not have symptoms but carries disease to their offspring.**
20. Gene patenting is related to **gene creation/ created gene.**
21. Microbes cause disease **WHO risk group.**
22. Neurodegenerative incurs in age **40.**

Subjective 2 marks

Qno1: Define Bio security.

Bio security is a set of preventive measures designed to reduce the risk of transmission of infectious agents. It is security against the inappropriate use of potentially dangerous biological agents.

Qno2: Name the two publications related to bioethics and also their Authors.

Potter "Bioethics, the survival of science and Callahan's "Bioethics as a discipline

Qno3: Write down the advancement of health and medicine in Biotechnology.

- Vaccinology
- Fiagnosis/ gene therapy
- Genetically modified embryos
- Xenotransplants
- Designer babies

Qno4: What are the ethical issues in treatments of older sibling?

Normal fertile couples undergo in vitro fertilization in order to produce a baby that can be a stem cell donor for an older sibling. The older sibling suffers from genetic disorder and the embryo created in vitro would be tested for the absence of mutation and is the positive tissue match to the older sibling.

The condition suffered by the older sibling is not genetic but the child still needs donated stem cells. In this case, in vitro embryo would be selected solely as a tissue match.

Qno5: What is the code for nurses for their coworker?

Nurses also have a good relationship with their co-worker. They have ability to teach ethics to their co-worker and respect them.

Qno6: How Biotechnology can be more miss use?

- Reality of war
- Inequitable distribution of resources
- Frequent misuse of science act as constant reminder that our actions do not always live up to our aspirations

Qno7: Write down the properties of Bio hazards symbol.

Developed in 1966 ---- Charles Baldwin

- Easy to sketch
- Labeling of biological materials
- Recognize quickly
- Acceptable to groups --ethnic backgrounds

Qno8: What is sentiency?

It is a capacity to explore pain and pleasure and capacity to experience satisfaction and dissatisfaction.

Qno9: Difference between Genetic screening and genetic testing.

Genetic screening can be defined as systemic search for person with specific genotype and individual or group show a risk of disease.

Genetic testing is specific or multiple gene interaction.

Qno10: Define Cybernetics.

Exploring regulatory system, their structures, functions. Greek word "governance". It is study of interactions between man, machine and animals.

Qno11: Write down two precautions to use the thermometer.

- Heat sources
- Avoid mouth pipetting
- Centrifuges
- Compressed gases

Qno12: What are patients 'rights and responsibilities for refusal treatment?

- Patient has a right to decline treatment
- Unethical to force the patient
- Patient must understand the consequences of refusal

Qno13: Write down the moral principle of dealing with human embryo.

- 1) Duty to prevent the suffering of the individual.
- 2) Respect the value of human life.

Qno14: How many types of Bio weapon?

Three types of Bio weapons

- 1) Chemical weapon
- 2) Radiological weapon
- 3) Nuclear weapon

Qno15: What type of mutation causes the Huntington disease?

Huntington disease is caused by the inherited defects in a single gene. This is caused by the point mutation in the genetic sequence.

Qno16: State personal health confidentially.

Patient has right to keep the information or the medical record confidential, because it is not the property of a hospital.

Qno1: How human activities cause loss of biodiversity?

- 1) Transformation of forests into lands
- 2) Pollution affects the ecosystem-loss of biodiversity
- 3) Tropical rain forest climax ecosystem

Qno2: Write down the three principles of Bioethics.

- Autonomy
- Nonmaleficence and beneficence
- Justice ----- example

Qno3: Write down the advantages of patenting a gene.

Advantages:

- Commercial interest
- Synthesis of generic drugs--AIDs
- Genetic based treatments
- Africa-- increased cost --gene patenting

Qno4: What are cybernetics ethical issues? Enlist any three.

1. Machines are in charge of key human functions
2. Wealthy ones can communicate through cybernetics
3. Implant is safe to use

Qno5: What are biological agents of bio weapons any six?

1. Anthrax
2. Brucellosis
3. Small pox
4. Viral hemorrhagic fever
5. Staphylococcal enterotoxin B
6. botulinum toxins

Qno6: Write down the advantages of UV sterilization.

- No known toxic residuals
- Short contact time
- Equipment is easy to operate and maintain

Qno7: Differentiate between sterilization and disinfectant.

Sterilization--removal of microbes including bacterial spores

Disinfection-killing many microbes

Qno8: Enlist the codes for nurses.

- Nurses and people
- Nurses and practice
- Nurses and profession
- Nurses and co-workers

Qno9: Why we use animals in Biotechnology?

Closer biochemical similarity to humans

Large amount of products

Qno10: What are design babies?

Children-genetically engineered in the womb to have desired qualities and made through in vitro fertilization embryo is removed manipulated for desired qualities placed in the womb.

Qno11: Explain Exploitation of victim in human trafficking?

This is very common in women and girls. It leads to prostitution and number of diseases transmission for example HIV, AIDS or hepatitis.

Qno12: Write down the aims of disaster management.

Aims:

- Assistance to maintain life
- Improve health
- Support the morale affected population
- Limited aid

Qno13: Write down the ten components of biological spill kit.

Household bleach • 70% alcohol • spray bottle • sterilization bags • absorbent paper towels • yellow trash bags • disposable gloves • metallic tongs • surgical masks • safety goggles, shoe covers and face mask • spill control and cleanup procedures.

Qno14: What is zoonosis also gives example?

Zoonosis is a term used for the diseases which is transmitted from animals to human are called zoonosis e.g Q fever.

VU Medical Zone Effort By Amaan Khan

household bleach, 70% alcohol, spray bottle, sterilization bags, absorbent paper towels, yellow trash bags, disposable gloves, metallic tongs, surgical masks, safety goggles, shoe covers and face mask, spill control and cleanup procedures.

14) What is the legal status of Fetus.

Topic No: 27

Ans: According to US Law fetus has rights, because it is a viable form. It has sound, body and mind. So charges of fetal abuse, refusal of hospitalization, intrauterine transfusion or surgical delivery is harmful for both mother and fetus.

15) What is zoonosis also give its example?

Topic No: 77

Ans: Zoonosis: Zoonosis is the term which is use for those diseases that can be transmitted from animals to human beings and these diseases are known as zoonotic diseases. For example Q fever can occur in cattle like goat, cow, sheep etc. This Q fever can cause nausea, headache, respiratory distress in animals. So if the workers, scientist and researchers work with such animals then they must be vaccinated against Q fever.

5 Marks questions:

1) Effects of global warming, any 5 with detail

Topic No: 13

Ans: Global warming is one of the major issues of today's problems. It is basically due to the rise in concentration of CO₂, methane and other greenhouse gasses into the environment. There are certain adverse effects of Global warming as mentioned bellow,

- 1). Due to the global warming earth climate zone is shifting.
- 2). Polar ice start melting with increase in temperature that may cause the increases in Sea level.
- 3). The rise in sea level leads to the danger of flood.
- 4) There are some microorganisms which are Methane producing. Increase in temperature speed up the metabolic rate of such type of bacteria.
- 5). Several species get extinct due to these methane producing bacteria.

2) Application of cybernetics.

Topic No: 33

Ans: There is a special software that is use in cybernetics have the ability to read signals from the nervous system to record them and condition a data in such a way that it can be retransmitted. There are numbers of application due to this software that can read the data of our nervous system.

- i Wooden limbs are totally replaced. These limbs are controlled by our nervous system and are used instead of wooden limbs. These are specially designed by cybernetics.
- ii Heart pace maker and artificial retinas are also designed by cybernetics.
- iii For blind and deaf people cybernetics also play an important role. They can feel or hear through ultrasonic waves just like bats.

- iv In the people that suffering from stroke condition, silicon chip is used that function like nerves. So we can easily replace the lost neuronal functions.
- v In our university ID card a silicon chip is used that carry all the information of the person.

3) Explain the case study 8 of Spleen removal with details.

Topic No: 61

Ans: **Case study 8:**

A man presents with symptoms representing cancer and as part of his treatment spleen should be removed. The pathology department uses it to establish a cultured cell line in order to study the rare cancer. The cell line performs so well that the scientists collaborate with Biotechnology Company to patent it. They start to earn royalties from other laboratories and organizations that wish to use the cell line. When patient find all this he was amazed, nobody has taken the consent from him nor he has been informed by anyone about these developments, analyze the ethical issues.

Reasons: In terms of medical ethics, removal of the spleen was an act of doing good-beneficence. Patient's personal autonomy had been respected. It was for the sake of his health that spleen has been removed. In UK, there is great sensitivity concerning the fate of organ removed. Under new legislation, if there is no pre-death consent then kin permission must be taken to retain any organs from dead bodies. In USA, once the organ is removed during surgery, it is no longer belongs to the patient. What is the purpose of keeping their appendix or diseased kidney in a jar in their office? Donor cannot claim on the income gained as a result of research, who donates a kidney makes a gift not an investment in the recipient.

4) Differentiate between intrinsic values and instrumental values.

Topic No: 09

Intrinsic value:

According to these values that environment and living forms have their own rights.

There are some intrinsic value and rights of birds/green and pleasant places. It is mainly involves religion.

Instrumental value:

Supply of human's towards the material needs such as natural resources. How human being take benefits from soil, water, air, minerals etc. So actual and potential use to supply those resources for humans, is very important.

5) Enlist four disciplines which show relationship of human with nature. Topic No: 08

Ans: There are four disciplines as mentioned below,

1. Anthropocentrism

This approach is centered to human beings. it is believe that rest of the nature is being there for the good of humans. Human greediness left ecological footprints that cause massive alteration in nature balance. E.g extensive use of water, electricity etc.

2. Ecocentrism:

it is Centered on ecosystem. Ecosystem includes soil, water, air, forest, mountains etc. Biotic

components are dependent on these non-livings. Such type of human ill treatment altering the nature that threatening for humans. Humans are responsible for all biological life because of having ability of thinking and perceiving world as a whole.

Rolston was the father of environmental ethics. He says, ecosystem is much more than the sum of its parts.

3. Biocentrism:

it is centered on biosphere. Here the theory of consequentialism is involved. Humans are one of many millions of species. What are results of bad and good conduct of humans, it doesn't prevent humans from using natural resources if other living organisms doesn't matter.

4. Theocentrism:

it is basically God centered approach to the world. It Comes from the religious faith. All religions believe that God is a creator of this universe. So environment belongs to God.

6) Define and explain Bio hazards classification?

Topic No: 70

Classification: we can characterize the biohazards into 4 groups.

- 1) Category A, UN 2814- Here the infectious substances affect the humans. They are fetal or life threatening for only human beings.
- 2) Category A, UN 2900-infectious substances affecting animals. They are fetal or life threatening for only animals.
- 3) Category B, UN 3373- Here we talk about the biological substances transported for diagnostic or any other investigative purposes.
- 4) Category UN 3291- It includes regulated medical waste. Here we talk about the waste or reusable materials which are derived from medical treatment/research. For example through testing or by production.

7) Write down the step of patent gene.

Topic No: 47

In 1st step we will isolate gene from rest of the DNA.

In 2nd step we will make a copy of it from mRNA population.

In 3rd step we will the turn gene sequence into an invention

In 4th step we will not use the gene itself but use a copy made in the test tube.

8) The element of human trafficking?

Topic No: 65

Elements: There are three basic elements of trafficking. the act (what is done), the means (how it is done), the purpose (why it is done)

1). Act: It can be in the form of recruitment, transport, transferring of individual, harboring, receipt of person.

2). Means: by which means organisms are transfer from one place to other. It can be in the form of threat, use of force, abduction, fraud and abuse of power or in the form of incentives. For example in the case of child labor payments/benefits are given to the family members.

3). Purpose: It leads to the range of **Exploitation**. This is very common in women and girls. It leads to prostitution, and number of diseases transmission e.g sexually transmitted like HIV, AIDS or hepatitis, so sexual exploitation is there. Others are forced labor, slavery, removal of organs and sell them at high prices.

9) Write down the WHO classification of pathogens.

Topic No: 73

In Pakistan WHO (World Health Organization) has classify the pathogens into four basic groups.

- 1). WHO risk group 1:** This group is related to the microbes unlikely to cause disease.
- 2). WHO risk group 2:** This group is related to microbes causing diseases but unlikely to be serious. Such as mild disease i.e Hepatitis A caused by a virus due to use of contaminated water.
- 3). WHO risk group 3:** This group is related to pathogens causing serious disease but they are easily treatable.
- 4). WHO risk group 4:** This group is related to bio weapons those pathogens which are causing serious disease that can be transmitted. There is no effective treatment or preventive measures for this. For example Dengue virus, Ebola virus etc.

10) Write down the codes of ethic of biologist.

Topic No: 18

Ans: Whenever scientists perform an experiment they should submit a synopsis in which they described their experiments and procedure. It is important that there should be a best interpretation of results. They have to summarize data honestly. At the end of article it is better to acknowledge the contributors for publications. They have to treat manuscript confidentially. They should not submit inaccurate or misleading information. They should disclose financial resources during the presentation or thesis. We have to help our colleagues and support professional organization.

11) Types of patients advance directives explain in detail.

Topic No: 24

Ans:Types: we have divided advance directives into two forms.

- 1) General:** It can be included donation of kidney, cornea of eye etc.
 - 2) Detailed:** When we want to make a treatment plan, it can be a detail legal document.
- There are further 2 types of advance directives.

- 1. Living will:** In this patient takes about treatment plan. E.g a kidney patient decide whether he wants to adopt the dialysis or not. It is limited to patient's will. In some countries living will acceptable orally or verbally. Another form of living will is terminal illness. That is irreversible. For example if patient die shortly then it will be more beneficial to know his living will.
- 2. Health care power of attorney:** It is also known as durable power of attorney. In this patient decides his own agent/proxy. This agent is responsible for making his health decisions.

12) How the genetic counselors design the session?

Topic No: 64

Session structure: it is divided into five phases.

- 1). **Intake phase:** } Both these phases take place outside the session.
2). **Follow-up phase:** }
- 3). **Encounter phase:** There is a dialog exchange between the genetic counselors and patient.
- 4). **Summary phase:** genetic counselors put all options in front of patient so that he can choose technique and methods for management and family planning.
- 5). **Initial contact:** it is a contact of genetic counselors with patient and his relatives.

13) Write down the ethical issue of treatment of animal in lab?

Topic No: 37

Ans:

- As moral values tells us right and wrong and ethical issues tells us reasons and judgment why we have taken wrong and right decision.
- When we talk about the ethical issues of animal treatment we use a term “sentiency”. It means capacity to explore pain or pleasures and comfort and discomfort etc.
- According to the term speciesism we agree that the animal suffer from pain but it is ethically significant.
- Peter was a philosopher he says that every animal has its own right. We should follow them either we use animal for any purpose like transportation, sports, shows and fashion etc.
- There is also a religious concern that God has made world and put everything in a natural order so through biotechnology system we get disturb it by transfer the genes to others. For example a number of ethical and religious issues arise by Mule production.

14) Write down the assessment of risks during Biotechnology practical involved DNA and gene.

Topic No: 30

Ans: Whenever we want to assess the risk factors we have to consider certain properties as mention below.

- 1). Source of DNA of the target gene/non-target DNA segment of the construct use.
- 2). The site of incorporation of the transgene within the recipient human genome.
- 3). What is the product of the transgene? and how it interacts with the environment and molecules of consumers.
- 4). We have to look at the pleiotropic effects of transgenic product and possible molecular changes that occur in transgenic products.
- 5). We also have to look tissue specificity for transgenic expression.
- 6). Numbers of transgenic organisms and how they are interacting with the environment

15) Heat two types in sterilization with example.

Topic No: 76

Ans: We have divided sterilization by heat into two major categories.

1). Moist heat:

- i. We sterilize the water by boiling at home.
- ii. Steaming is another way of sterilization.
- iii. In labs we use autoclave to sterilize glass apparatus and solutions at 121°C and pressure should be of 15 for about 15-20 minutes.
- iv. Pasteurization is the process by which we sterilize milk below 100°C. We have to destroy all microorganisms from milk and side by side maintain its constituents.

2). Dry heat:

- i. By red heat: If we are using inoculating loop in micro biological lab then we have to sterilize it before using. Simply we put it on the blue flame of the fire and when it gets red hot it will sterilize.
- ii. By flaming we sterilize glass rod. For this simply 1st we have to dip it in the spirit and then pass it through the flame.
- iii. We can also sterilize glass apparatus by placing it in hot air oven at 80°C.

16) What is low risk in TB lab? Explain.

Topic No: 87

Ans: **Low risk TB labs:**

It has minimum bio safety requirements so it is called low risk TB Lab. The 1st step of identification of mycobacterium tuberculosis is microscopy. For this purpose we use ziehl-neelsen stain.

- Smear preparation: During smear preparation there should be less chances of aerosols production. For this we use direct sputum because sputum is a semi viscous substance that has less chance to produce aerosols. Two main reasons are there for less contamination. First we have to air dry the smear and second we go through the fixation process to kill the bacteria.
- Preparation of specimen: If we want to purify DNA as before performing polymerase chain reaction we have to prepare our specimen for automated nucleic acid purification assay. In this case there also should be less chances of aerosols production.